EXPERIMENT 2: STUDY OF NETWORK CABLES

AIM

To study different types of network cables and construct straight-through and crossover Ethernet cables using a crimping tool.

A. TYPES OF NETWORK CABLES

1. Unshielded Twisted Pair (UTP): Consists of twisted copper wire pairs without additional shielding. It is inexpensive and easy to install, but is more susceptible to electromagnetic interference. It is commonly used in Ethernet LANs.

2. Shielded Twisted Pair (STP): Consists of twisted pairs with shielding to reduce interference. It provides better noise protection but is more expensive and requires careful installation.

3. Coaxial Cable: Contains a central conductor, insulating layer, metallic shield and outer jacket. It is used in cable television and some broadband connections.

4. Fibre-Optic Cable: Transmits data as light through optical fibres. It supports high bandwidth and long-distance communication and is immune to electromagnetic interference.

B. STRAIGHT-THROUGH AND CROSSOVER CABLES

Straight-through cable: Both ends use the same wiring standard (T568A–T568A or T568B–T568B). It is traditionally used to connect unlike devices, such as a PC to a switch.

Crossover cable: One end uses T568A and the other uses T568B, crossing the transmit and receive pairs. It is traditionally used to connect similar devices directly when auto-MDI-X is unavailable.

TOOLS REQUIRED

UTP Ethernet cable (e.g., Cat5e), two RJ-45 plugs, wire stripper/cutter, crimping tool and a cable tester (if available).

PROCEDURE

1. Cut the Ethernet cable to the required length. Slide protective boots onto the cable first, if provided.

2. Strip approximately 1.5 cm of the outer jacket at both ends without damaging the inner wires.

3. Untwist and arrange the eight wires in the required order.

4. For a straight-through cable, use the same wiring standard at both ends. For a crossover cable, use T568A at one end and T568B at the other.

5. Trim the wires evenly and insert them fully into the RJ-45 plug, ensuring that the outer jacket reaches under the strain-relief tab.

6. Crimp the plug using the crimping tool. Repeat at the other end using the selected wiring order.

7. Test the cable using a cable tester or connect compatible devices and check the link.

WIRING ORDER (PIN 1 TO PIN 8)

T568A	T568B
White-green Green White-orange Blue White-blue Orange White-brown Brown	White-orange Orange White-green Blue White-blue Green White-brown Brown

OBSERVATION

The wires were arranged according to the selected standard and crimped into RJ-45 plugs. A correctly made cable should show the expected pin mapping on a cable tester and establish a link when connected to compatible devices. The actual tester indication or link status should be recorded after performing the experiment.

RESULT

Thus, the major types of network cables were studied, and the wiring arrangements and construction of straight-through and crossover Ethernet cables were understood.